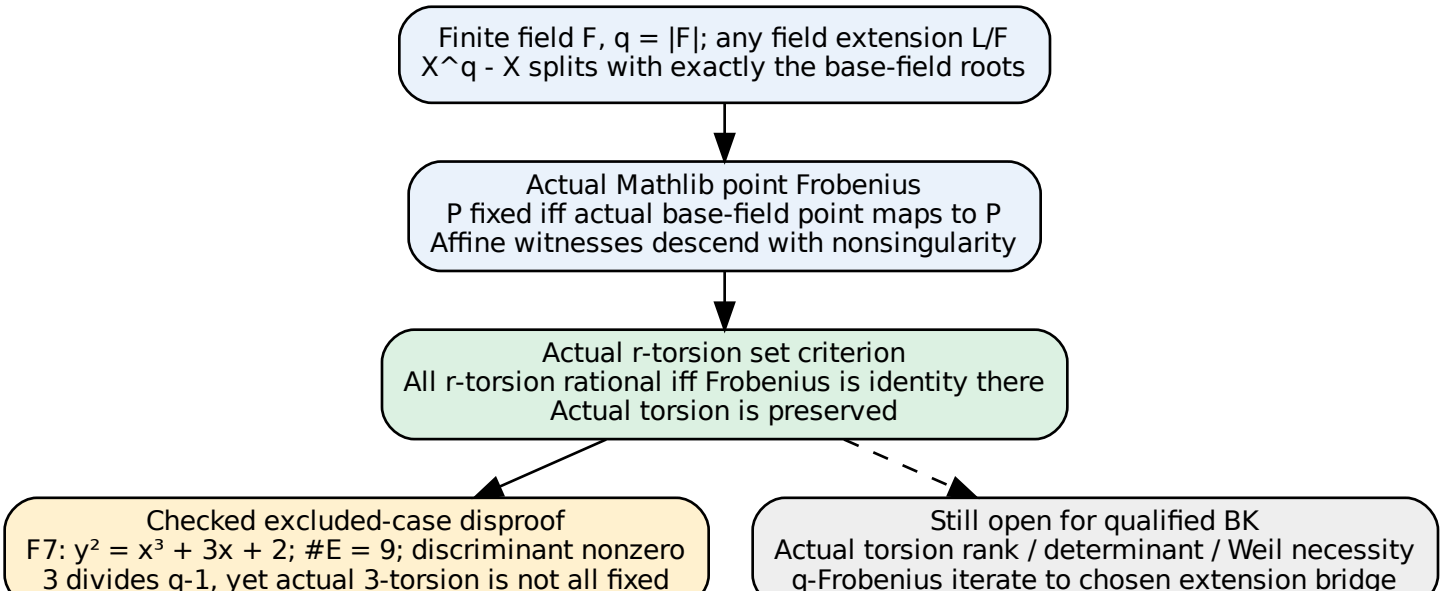
Finite field F, q = |F|; any field extension L/F
X^q - X splits with exactly the base-field roots
Actual Mathlib point Frobenius
P fixed iff actual base-field point maps to P
Affine witnesses descend with nonsingularity
Actual r-torsion set criterion
All r-torsion rational iff Frobenius is identity there
Actual torsion is preserved
Checked excluded-case disproof
F7: y² = x³ + 3x + 2; #E = 9; discriminant nonzero
3 divides q-1, yet actual 3-torsion is not all fixed
Still open for qualified BK
Actual torsion rank / determinant / Weil necessity
q-Frobenius iterate to chosen extension bridge